



CFE Information Circular IC05-07

April 21, 2005

To: All Volatility Index Market Participants

Re: CBOE DJIA Volatility Index and CBOE VIX Futures Settlement

In preparation for the listing of CFE's newest volatility index futures contract, the CBOE DJIA Volatility Index (VXD) futures contract, CFE is issuing this circular to remind market participants of the features inherent in the trading of volatility index futures contracts. This information circular updates CFE Information Circulars IC04-06, IC-04-07, IC04-08, and IC05-04 and summarizes information previously provided in those circulars.

I. Settlement Procedure for Volatility Index Futures Contracts

All volatility index futures contracts settle on the Wednesday prior to expiration Friday that is thirty days prior to the third Friday of the calendar month immediately following the month in which the applicable volatility index futures contract expires. This means, for example, that the June 2005 VXD and CBOE Volatility Index (VIX) futures contracts will settle on Wednesday, June 15, 2005, which is thirty days prior to the settlement date of the corresponding July 2005 options on the Dow Jones Industrial Average (DJX) and on Standard & Poor's 500 Stock Index (SPX), respectively (Friday, July 15, 2005). The Final Settlement Value for volatility index futures traded on CFE (including VXD and VIX futures) is a Special Opening Quotation (SOQ) of the volatility index calculated from the sequence of opening prices of the constituent options used to calculate the volatility index on the settlement date (Constituent Options). The opening price for any constituent options series in which there is no trade will be the average of that option's bid price and ask price as determined at the opening of trading.

The opening prices for the Constituent Options used in calculating the SOQ are determined through an automated auction mechanism (Rapid Opening System or ROS) that matches buy and sell orders residing on the electronic order book prior to 8:28 a.m. (Chicago time). Market participants should review CBOE Regulatory Circulars RG04-85 and RG05-41 for information regarding the opening procedures on CBOE for the Constituent Options used to calculate volatility indexes.

II. Settlement Methodology for Volatility Index Futures Contracts

The CBOE Research Department calculates the SOQ for volatility indexes using the following procedure:

- A. Collect the following information for each eligible option series of the applicable volatility index from CBOE's Time & Sales:
 - (i) The opening traded price, if any; and
 - (ii) The first bid/ask quote for each eligible option series of the applicable volatility index futures.
- B. Determine the forward index level, F , for each eligible contract month based on at-the-money option prices. The at-the-money strike is the strike price at which the difference between the call and put mid-quote prices is smallest.
- C. Determine K_0 - the strike price immediately below the forward index level.

- D. Sort all of the options in ascending order by strike price. Select call options that have strike prices greater than K_0 and a non-zero bid price, beginning with the strike price closest to K_0 and moving to the next higher strike prices in succession. After encountering two consecutive calls with a bid price of zero, do not select any other calls. Next, select put options that have strike prices less than K_0 and a non-zero bid price, beginning with the strike price closest to K_0 and then moving to the next lower strike prices in succession. After encountering two consecutive puts with a bid price of zero, do not select any other puts. Select both the put and call with strike price K_0 .
- E. Calculate volatility index SOQ using the options selected. The price of each option used in the calculation is the opening traded price of that option. In the event that there is no opening traded price for an option, the price used in the calculation is the average of the first bid/ask quote for that option.
- F. Multiply the volatility index SOQ by ten in order to determine the Final Settlement Value for the applicable volatility index futures contract.

III. Risk Inherent in Settlement Procedure

The Final Settlement Value of each volatility index contract is calculated from the actual opening prices of the Constituent Options on the final settlement date. In contrast, all other volatility index values disseminated during the life of volatility index futures contracts are "indicative" values – namely, values that are calculated using the mid-point of the disseminated bid and offer quotations of each of the Constituent Options at a particular time. These indicative values are published as VXB for VIX futures and DVB for VXD futures and are equal to ten times the indicative values of the applicable volatility index.

Because actual prices are used to compute the Final Settlement Value of volatility index futures while mid-market options quotes are used to compute indicative volatility index values, there is an inherent risk of a significant disparity between the Final Settlement Value of an expiring volatility index futures contract and the opening indicative volatility index value on the final settlement date. In fact, such disparities have occurred in the past with respect to VIX futures contracts. Accordingly, because volatility index futures settle based on the trade price of the Constituent Options established during the ROS opening, rather than on quotes, investors should be aware that the possibility exists, as occurred in the past, that there could be a significant difference between the Final Settlement Value for a volatility index futures contract and the previous day's closing indicative volatility value, or the opening indicative volatility value on the final settlement date.

For example, one type of hedge for VIX futures involves holding a portfolio of the SPX options that will be used to calculate VIX on the final settlement date. Traders holding hedged VIX futures positions to settlement can be expected to trade out of their SPX options on that date. Traders who hold short, hedged VIX futures would liquidate that hedge by selling their SPX options, while traders holding long, hedged VIX positions would liquidate their hedge by buying SPX options. In order to seek convergence with the VIX Final Settlement Value, these traders would be expected to liquidate their hedges by submitting orders in the appropriate SPX option series during the SPX opening on the final settlement date of the VIX futures contract. To the extent (1) traders who are liquidating hedges predominately are on one side of the market (*e.g.*, seek to buy the particular SPX options) and (2) those traders' orders predominate over other orders during the SPX opening on the final settlement date for the VIX futures contract, trades to liquidate hedges may contribute to an order imbalance during the SPX opening on that date. If there are order imbalances significantly weighted on the same side of the market in SPX option series used in the final settlement, there will be a disparity between the Final Settlement Value and the VXB index values that are reported after the SPX opening. In fact, these factors did occur,

and did lead to this type of imbalance and disparity, during past SPX openings for VIX futures settlement days. The same is equally applicable with respect to VXD futures.

In order to avoid exposure to such disparities, investors in volatility index futures holding speculative or un-hedged volatility futures may wish to either close out their positions or roll to another contract month prior to settlement.

IV. Improvements to the Settlement Process for Volatility Index Futures:

CFE has previously taken the following steps to improve the settlement process for volatility index futures:

- CFE disseminates updated indicative values for volatility indexes based largely on option bid and offer prices four times per minute throughout the trading day (8:30 – 3:15 p.m. (Chicago time)). The indicative values are disseminated via the CBOE Financial Network and CFE via Bloomberg and ILX. The indicative values provide market participants with more information on the possible range of settlement prices that may occur for volatility index futures contracts. However, market participants should be aware that the Final Settlement Value may be outside of the previous day's range of the applicable indicative values.

Symbols for CBOE Volatility Index Indicative Values (VXB):

VWB: based on SPX option bid prices.

VWA: based on SPX option offer prices.

Symbols for CBOE DJIA Volatility Index Indicative Values (DVB):

DDB: based on DJX option bid prices.

DDA: based on DJX option offer prices.

Note: CFE had previously published the VXB bid and VXB offer values on its web pages once per day at 2:30 p.m. (Chicago time) only during the week prior to VIX futures contract settlement. This practice has been discontinued since these values are now disseminated via the CBOE Financial Network.

- Volatility Index Futures Roll-Period Information: To promote awareness by market participants of the potential roll market, certain CFE Trading Privilege Holders have agreed to quote the net price to buy (or sell) the volatility index futures (including VIX and VXD futures) for the current contract month and simultaneously to sell (or buy) the volatility index futures for the next contract month. CFE will publish these indicative roll quotes once per day at 2:30 p.m. (Chicago time) on the four business days preceding the volatility index settlement day. This information will be located on the CFE website at www.cboe.com/cfe. Because the actual price to complete a roll transaction will change over time in response to market conditions, the indicative roll market that is published once per day during the four-day roll period will be for informational purposes only and will not represent firm prices at which a market participant will be able to trade.
- CFE is continuing to work on improvements to the settlement procedures for volatility index futures contracts and will notify market participants of any such changes prior to implementation.

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